

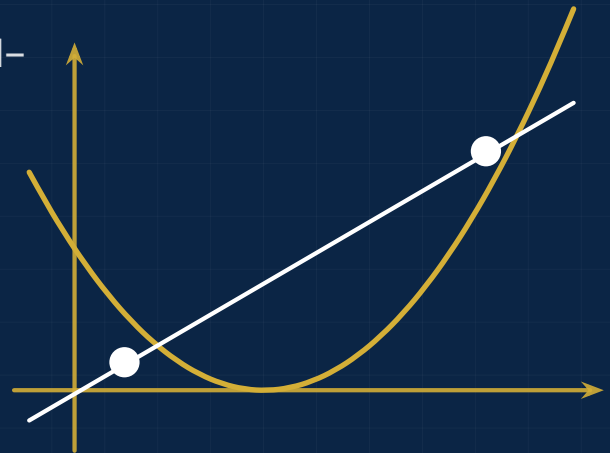


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CHAPTER 3 • SEQUENCES, FUNCTIONS & GRAPHS • TOPIC 7

Real-Life Graphs

Read the axes first. On a distance–time graph the gradient is speed; on a speed–time graph the gradient is acceleration and the area is distance.



4

VARIANTS

0

PAST PAPERS

35

PRACTICE QUESTIONS

YOU WILL MASTER

V1 Distance–time

V2 Speed–time: acceleration

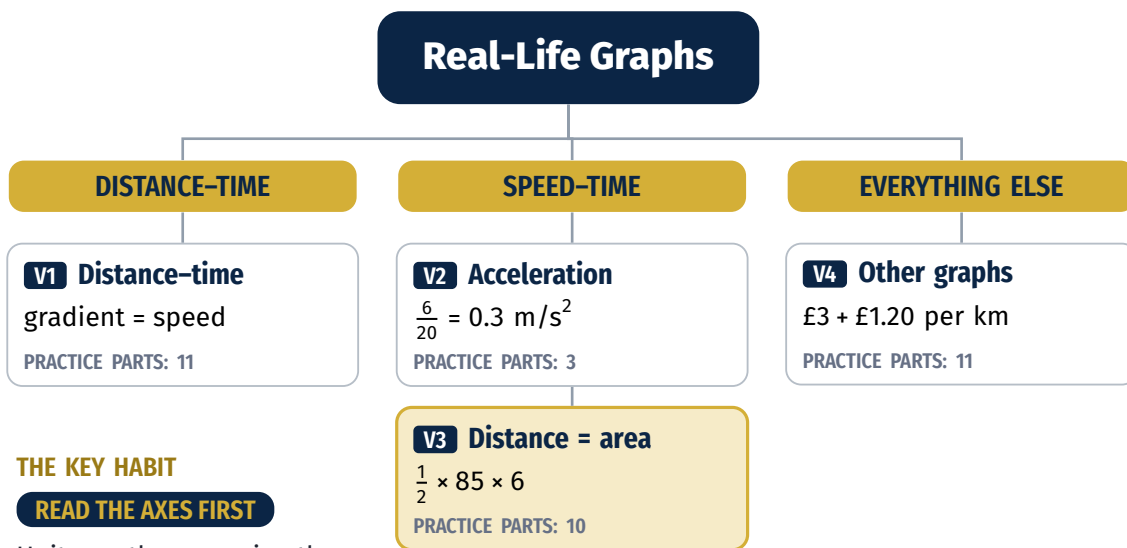
V3 Speed–time: distance

V4 Other real-life graphs

TOPIC MAP • EVERY VARIANT ON ONE PAGE

Real-life graphs at a glance



Your notes
**THE KEY HABIT****READ THE AXES FIRST**

Units on the axes give the units of every answer.

★ How often does each variant come up?

No past-paper question is only a real-life graph — they appear inside gradient, area and proportion questions. 35 practice parts, grouped by type.

	PRACTICE PARTS
V1 Distance-time	11
V2 Speed-time: acceleration	3
V3 Speed-time: distance	10
V4 Other real-life graphs	11

WHAT THIS TELLS YOU These skills are tested inside other questions (Estimating Gradients, Areas, Kinematics in M1). Know what the gradient and the area mean on each graph and you will recognise them anywhere.

BEFORE YOU START

The big idea

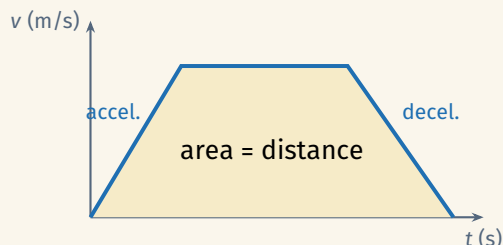


Gradient and area mean something.

Gradient = $\frac{y \text{ units}}{x \text{ units}}$ — a rate. **Area** under the graph = $(y \text{ units}) \times (x \text{ units})$ — a total.

Distance–time: gradient = speed.

Speed–time: gradient = acceleration,
area = distance.



What each feature means

Flat (D–T)

stopped

Distance not
changing.

Flat (S–T)

steady speed

Acceleration 0.

Steeper

faster

Bigger gradient,
bigger rate.

Going down

back / slowing

D–T: returning.
S–T: decelerating.

Units

km and minutes

$$20 \text{ min} = \frac{1}{3} \text{ h}$$

Change minutes to hours
for km/h.

m/s to km/h

$$\times 3.6$$

$$6 \text{ m/s} = 21.6 \text{ km/h.}$$

km to m

$$1.75 \text{ km} = 1750 \text{ m}$$

Match the graph's units.

**CALCULATOR FIRST, THEN CHECK**

Speed = $\frac{\text{distance}}{\text{time}}$ — write the triangle and the units on every line.

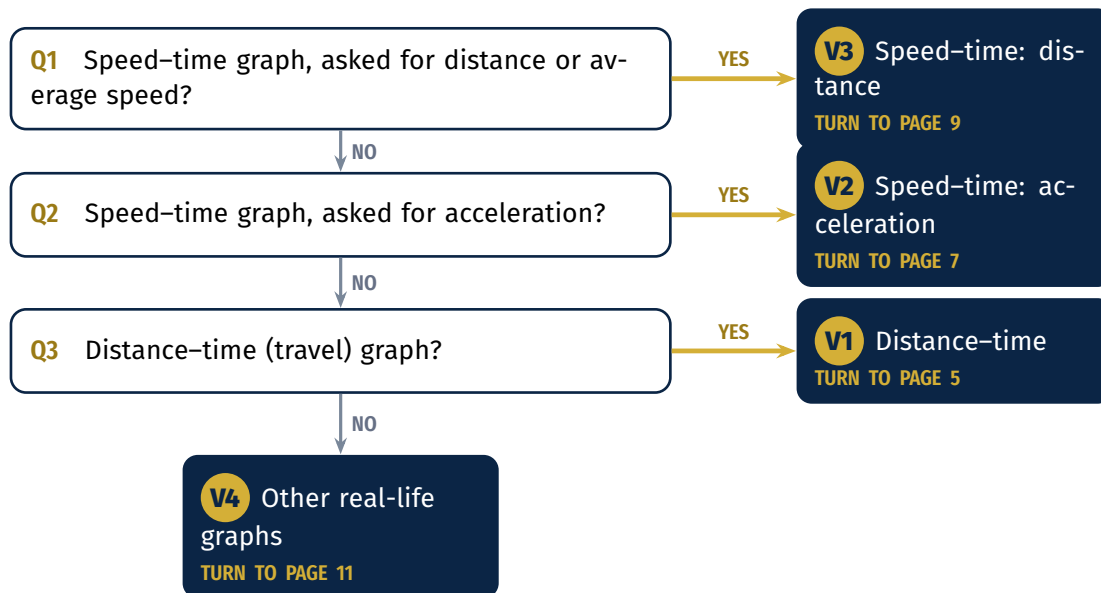
COUNT / RECOGNISE, THEN CHOOSE

Which variant am I?



Your notes

Read the question, then walk down the ladder. The first **YES** tells you the variant — and the page to turn to.



COLOUR CODE IN THE WORKING

Blue = letter terms $4x$, $-3y$

Orange = plain numbers 17 , -13

STEP 1 **STEP 2** **STEP 3** **STEP 4** = which step of "How do I do it?" each line is.

ON EVERY VARIANT SPREAD



What is it?



When you see → you do



How do I do it?



Watch out



Worked example



Exam tip



Check yourself



Calculator check

V1**VARIANT 1**

Distance–Time Graphs

Gradient = speed; flat = stopped; back down = returning.

PRACTICE TOPIC
NO PAST-PAPER QUESTION OF ITS OWN
USED INSIDE OTHER QUESTIONS
PRACTICE QUESTIONS: 11


Your notes

? What is it?

In plain words

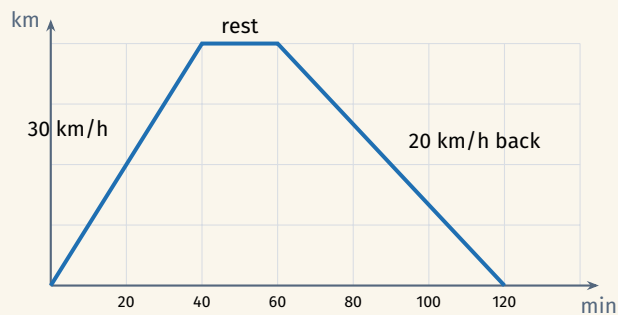
Each straight section is a part of the journey at a **steady speed**: speed = **distance ÷ time** for that section.

A **flat** section means stopped; a line going **down** means coming back.

IT LOOKS LIKE

speed of each section

draw the return



How do I do it?

1 Section Pick one straight section.

2 Read Distance change and time taken.

3 Hours Minutes ÷ 60 for km/h.

4 Divide Speed = distance ÷ time.

STEEPER = FASTER Flat = stopped.

V1 Distance-Time Graphs

VARIANT 1 • SEE IT DONE, THEN TRY

 Your notes
WHEN YOU SEE → YOU DO

"Work out the speed ... between 2 pm and 2:40 pm" → **Distance ÷ time in hours**

"During which part was she fastest?" → **Steepest section**

"She rests for 20 minutes, then returns at ..." → **Flat line, then a line back to 0**

WORKED EXAMPLE DR ESAM

Distance-time • 3 marks

Sam cycles 20 km from home in 40 minutes, rests for 20 minutes, then cycles home in 60 minutes. Find his speed on the way out and on the way back, in km/h.

STEP 1 $40 \text{ min} = \frac{2}{3} \text{ h}$ ◀ minutes to hours

STEP 2 $20 \div \frac{2}{3} = 30 \text{ km/h}$ ◀ out

STEP 3 $60 \text{ min} = 1 \text{ h}$ ◀ back

STEP 4 $20 \div 1 = 20 \text{ km/h}$ ◀ back

ANSWER out 30 km/h, back 20 km/h ✓ out section is steeper ✓

WATCH OUT

✗ $20 \div 40 = 0.5 \text{ km/h}$

✓ $20 \div \frac{2}{3} = 30 \text{ km/h}$

Minutes must become hours for km/h.

EXAM TIP

Write the time as a fraction of an hour:

$$24 \text{ min} = \frac{24}{60} = 0.4 \text{ h.}$$

Check yourself

calculator first • answers on page 13

1.1 Dr Esam

A cyclist travels 18 km in 45 minutes. Work out the average speed in km/h.

Answer:

1.2 Practice • Medium

A coach travels 160 km in 2.5 hours. Average speed?

Answer:

1.3 Practice • Medium

Debbie drives 30 km in 24 minutes; Ann averages 66 km/h. Who is faster?

Answer:

V2

VARIANT 2

**Speed–Time:
Acceleration**

Acceleration = gradient = change in speed ÷ time. Going down: deceleration.

PRACTICE TOPIC
NO PAST-PAPER QUESTION OF ITS OWN
USED INSIDE OTHER QUESTIONS
PRACTICE QUESTIONS: 3


Your notes
? What is it?**In plain words**

On a speed–time graph the gradient is the **acceleration**, in m/s^2 .

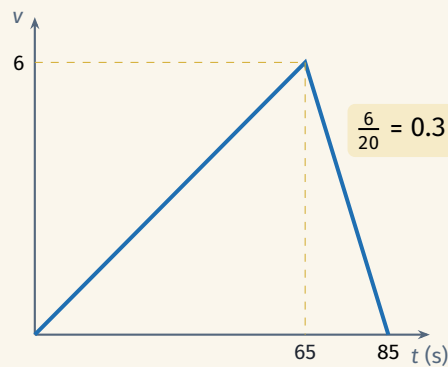
Flat: **constant speed**, acceleration 0.

Going down: **deceleration** (a negative gradient – give the size as a deceleration).

IT LOOKS LIKE

deceleration as it stops

when is acceleration 0?

**How do I do it?****1 Section**

The straight part asked about.

2 Change

Speed at the end – speed at the start.

3 Time

How long that part lasts.

4 Divide

$\text{m/s} \div \text{s} = \text{m/s}^2$.

GRADIENT = ACCELERATION Units m/s^2 .

V2 Speed–Time: Acceleration

VARIANT 2 • SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

“Calculate the deceleration as it approaches ...”

→ **Size of the downhill gradient**

“After how many seconds was the acceleration zero?”

→ **Where the graph is flat (top of a curve)**

“Describe the acceleration for each part”

→ **Constant / zero / deceleration, with values**

**WORKED EXAMPLE** **PRACTICE**

Practice • Medium • Q13(a) • 2 marks

A tram’s speed rises steadily from 0 to 6 m/s in 65 s, then falls steadily to 0 at 85 s. Calculate the deceleration as it approaches the town hall.

STEP 1 65 s → 85 s

◀ the last section

STEP 2+3 change = 6 m/s, time = 20 s

◀ 6 to 0; 85 – 65

STEP 4 $6 \div 20 = 0.3$

◀ gradient size

ANSWER 0.3 m/s²

✓ steeper than the first part ($\frac{6}{65}$) ✓

**WATCH OUT**

× deceleration = -0.3 m/s^2

✓ deceleration 0.3 m/s^2

“Deceleration” already means slowing — give the size.

**EXAM TIP**

Flat on a speed–time graph is not “stopped” — it is steady speed.

**Check yourself**

calculator first • answers on page 13

2.1

Dr Eslam

The speed rises steadily from 4 m/s to 16 m/s in 6 s. Find the acceleration.

Answer:

2.2

Practice • Easy

The speed–time graph is flat for 20 s. What is the acceleration?

Answer:

2.3

Practice • Hard

Final 10 s: speed falls steadily from 30 m/s to 0. Describe the acceleration.

Answer:



VARIANT 3

Speed–Time: Distance = Area

Split the area into triangles and rectangles (or one trapezium); the total is the distance.

PRACTICE TOPIC
NO PAST-PAPER QUESTION OF ITS OWN
USED INSIDE OTHER QUESTIONS
PRACTICE QUESTIONS: 10



Your notes

What is it?

In plain words

Distance = **area under** the speed-time graph. Split it into **triangles, rectangles** or one **trapezium**.

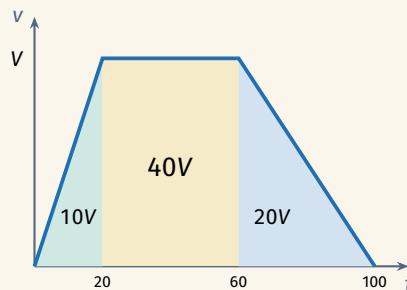
Average speed = total distance $\div$ total time. Unknown speed V ? Write the area in terms of V and solve.

IT LOOKS LIKE

distance travelled

find V

average speed



How do I do it?

1 Split

Triangles and rectangles under the graph.

2 Areas

 $\frac{1}{2}bh$ and bh (or $\frac{1}{2}(a+b)h$).

3 Add

Total = distance (check units).

4 Use

Average speed, unknown V , or a time.

AREA = DISTANCE The most common speed–time question.

V3 Speed–Time: Distance = Area

VARIANT 3 • SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

“Calculate the distance travelled” → **Area under the graph**

“The car travelled 1.75 km. Work out V ” → **Area in terms of $V = 1750$**

“Work out the average speed” → **Total area ÷ total time**

**WORKED EXAMPLE****PRACTICE**

Practice • Medium • Q13(b)(c) • 7 marks

The tram's speed rises steadily from 0 to 6 m/s in 65 s, then falls to 0 at 85 s. (b) Calculate the distance travelled. (c) Give the maximum speed in km/h.

STEP 1 one triangle: base 85, height 6 ◀ the shape

STEP 2 $\frac{1}{2} \times 85 \times 6 = 255$ ◀ (b) area

STEP 3 6 m/s $\times$ 3.6 ◀ (c) m/s to km/h

STEP 4 = 21.6 km/h ◀ convert

ANSWER (b) 255 m (c) 21.6 km/h

✓ $6 \times 3600 \div 1000 = 21.6$ ✓

**WATCH OUT**

✗ distance = 85×6

✓ $\frac{1}{2} \times 85 \times 6$

The shape is a triangle, not a rectangle.

**EXAM TIP**

Write each piece's area in the shape on the graph, then add them.

**Check yourself**

calculator first • answers on page 13

3.1

Practice • Hard

Up to V m/s in 20 s, steady to 60 s, down to 0 at 100 s. Distance 1.75 km. Find V .

Answer:

3.2

Practice • Hard

Steady k m/s for 40 s, then down to 0 at 50 s. Distance 1.35 km. Find k .

Answer:

3.3

Practice • Hard

A horse speeds up from 0 to 10 m/s in 4 s, then runs at 10 m/s. When has it run 75 m?

Answer:

V4**VARIANT 4**

Other Real-Life Graphs

Intercept = fixed amount; gradient = rate per unit; where two lines cross they are equal.

PRACTICE TOPIC
NO PAST-PAPER QUESTION OF ITS OWN
USED INSIDE OTHER QUESTIONS
PRACTICE QUESTIONS: 11

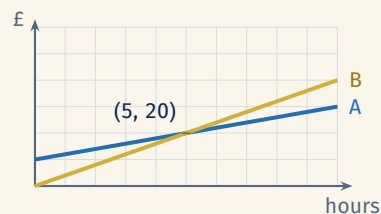

Your notes

? What is it?

In plain words

Costs, conversions, depths, temperatures: a straight line $y = mx + c$ where c is the **starting** (fixed) amount and m is the **rate** (per km, per hour, per second).

Two plans: the crossing point is where they **cost the same**.



IT LOOKS LIKE

taxi fares £ to €
depth of water

How do I do it?

1 Axes

What is on each axis, and the units.

2 Intercept

The fixed / starting amount.

3 Gradient

The rate: units of y per unit of x .

4 Compare

Crossing point: equal. Above / below: more / less.

READ THE AXES FIRST Then m and c tell the story.

V4 Other Real-Life Graphs

VARIANT 4 • SEE IT DONE, THEN TRY

 Your notes

WHEN YOU SEE → YOU DO

"What does the gradient represent?" → **Rate: ... per ... (with units)**

"What does the intercept represent?" → **The fixed charge / starting value**

"For what distance do they cost the same?" → **Where the lines cross**

WORKED EXAMPLE DR ESAM

Real-life graph • 3 marks

A taxi costs £3 plus £1.20 per km. (a) What do the gradient and intercept of its cost graph mean? (b) Find the cost of an 8 km trip.

STEP 1 $C = 1.2d + 3$ ◀ the line

STEP 2 $m = 1.2$: £1.20 per km ◀ (a) gradient

STEP 3 $c = 3$: fixed charge £3 ◀ (a) intercept

STEP 4 $1.2 \times 8 + 3 = 12.60$ ◀ (b)

ANSWER (b) £12.60

✓ $9.60 + 3 = 12.60$ ✓

WATCH OUT

✗ "the gradient is 1.2"

✓ "£1.20 for each km"

Say what it means, with units.

EXAM TIP

For two plans, solve the two equations equal — the graph reading is then a check.

Check yourself

calculator first • answers on page 13

4.1 Practice • Easy

A depth–time graph has gradient -1.5 (cm, seconds). What does it mean?

Answer:

4.2 Dr Eslam

A straight conversion line through $(0, 0)$ shows $£50 = €58$. Convert £120.

Answer:

4.3 Dr Eslam

Plan A: £10 + £2 per hour. Plan B: £4 per hour. When do they cost the same?

Answer:

CHECK YOURSELF

Answers



Every answer below was checked with sympy (check_answers.py).

1.1 24 km/h

1.2 64 km/h

1.3 Debbie: 75 km/h

2.1 2 m/s^2

2.2 0

2.3 deceleration 3 m/s^2

3.1 $V = 25$

3.2 $k = 30$

3.3 9.5 s

4.1 falls 1.5 cm each second

4.2 €139.20

4.3 5 hours (£20)

THE WHOLE TOPIC ON ONE PAGE

Real-life graphs cheat sheet



Your notes
V1 Distance–time

$$20 \div \frac{2}{3} = 30$$

Gradient = speed; minutes to hours.

V2 Acceleration

$$\frac{6}{20} = 0.3 \text{ m/s}^2$$

Gradient of speed–time.

V3 Distance

$$\frac{1}{2} \times 85 \times 6 = 255$$

Area under speed–time.

V4 Other graphs

$$C = 1.2d + 3$$

 c fixed, m rate, crossing equal.**TOP 3 LOST MARKS**

1. minutes used as hours ×
2. rectangle area for a triangle ×
3. gradient with no meaning ×

BOTTOM LINE Read the axes · gradient = rate · area = total · answer with units.